

DLD (EE1005)

Date: May 20th 2024

Course Instructor(s)

Mr. Kamran Jamal

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Final Exam

Total Time (Hrs): 3

Total Marks: 100

Total Questions: 4

Roll No

Section

Student Signature

Attempt all the questions. All questions carry equal marks and equal weights.

CLO #2: Apply Boolean Algebra and K-map methods to optimize logic circuits

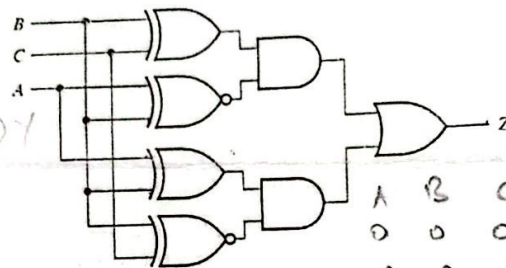
Q1:

Consider the following circuit

$$(B \oplus C)(\overline{A \oplus B}) + (A \oplus B)(\overline{B \oplus C})$$

$$x \quad y \quad + \quad x \quad z \quad = \quad x \oplus y$$

$$(A \oplus B) \oplus (B \oplus C)$$



A	B	C
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

✓ a) Construct the truth table for the function Z. SoP

✓ b) Write the function Z in canonical sum of product form (little m notation).

✓ c) Simplify the function to the minimum number of literals.

CLO #3: Design combinational circuits using functional blocks

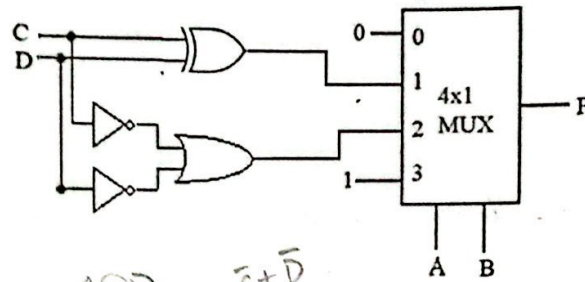
Q2:

- a) Design a combinational multiplier that has two 2-bit inputs A_1A_0 and B_1B_0 and 4 bit output $F_3F_2F_1F_0$. Construct the truth table and implement it using 4 to 16 line decoder and gates of your choice.

$$\begin{matrix} 8 & 4 & 2 & 1 \\ 2^3 & 2^2 & 2^1 & 2^0 \end{matrix}$$

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b) Construct the Truth table and write F in Σm notation.



CLO #4: Analyze elements of sequential circuits and design finite state machines

Q3:

a) Analyze a sequential circuit with three D flip-flops (A, B, C) defined by the following next state equations

$$\begin{aligned} D_A &= B\bar{C} \\ D_B &= C \\ D_C &= \bar{B} + X \end{aligned}$$

Where A, B and C are the outputs of the flip-flops A, B and C respectively, X is the external input and D_A , D_B and D_C are the input of flip-flops A, B and C respectively. Construct the state table and state diagram of this sequential circuit.

b) Consider a sequence recognizer circuit that has a serial input S and one output Z. The output becomes one when a sequence 1001 is detected and zero otherwise. Generate a state diagram only. Circuit implementation is not required.

CLO #5: Design of sequential circuits for various applications

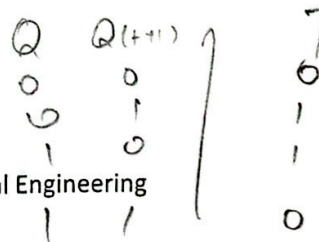
Q4:

Design a synchronous binary arbitrary counter using T Flip-Flop to produce the sequence 1,4,7,3,1 and repeat.

a) Construct the state diagram and state table for the required circuit.

b) Construct the logic diagram of your design using T-flip flops and logic gates.

c) Include the unused states in the state diagram and explain if this counter has self correction capability.



Spring 2024

Department of Electrical Engineering

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